assigns PA9 as TX and PA10 as RX. Set the baud rate to 9600.

Step 3: Configure I2C1. In the configuration panel, go to Connectivity, select I2C1, and set the mode to I2C. This automatically assigns PB6 as SCL and PB7 as SDA.

Step 4: Configure GPIO pins. In the chip diagram, select PA4, PA5, PA6, and PA7 one by one and configure each as GPIO Output. These pins control the IN1 to IN4 inputs of the motor driver.

Step 5: Generate the code. Press Ctrl + S to generate all the system files.

Step 6: Add the OLED driver files. Right-click core/src and add ssd1306.c and fonts.c. Then, right-click core/inc and add ssd1306.h and fonts.h.

Step 7: Build and upload. Click the build icon to compile the system. Connect the ST-Link to the Blue Pill using the SWCLK and SWDIO pins. Then, click Run or Debug to upload the code to the microcontroller.

Robot chassis design

The frame of the Bluetooth-controlled robot using the STM32 Blue Pill was designed using Autodesk Fusion 360 with a top-down parametric assembly approach, ensuring that the robot is structurally strong and suitable for manufacture using fused deposition modelling 3D printing.

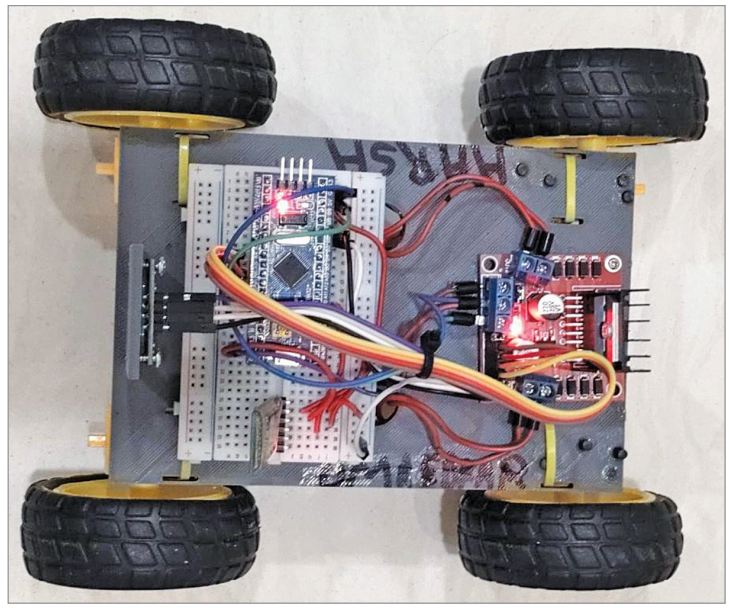


Fig. 6: Prototype of robot assembled on chassis

Fig. 5 shows the 3D-printed robot chassis, which serves as the central mounting plate for all the components. It measures 150mm in length, 100mm in width, and 5mm in thickness, providing sufficient strength to withstand the forces generated by the motors under load. The flat bottom surface simplifies the printing process by eliminating the need for support structures, thereby reducing print time and improving bed adhesion.

The main plate includes circular cut-outs to reduce weight and facilitate internal wire routing, along with peripheral slots for aligning and securing the motor brackets. A vertical bracket on the front edge holds a 0.96-inch (2.4cm) SSD1306 OLED display, which shows real-time status information based on the Bluetooth movement commands received by the STM32 Blue Pill microcontroller. Fig. 6 shows the prototype of the Bluetooth-controlled robot.

Construction and testing

After completing the chassis design, the source code is uploaded to the STM32 Blue Pill board and the circuit is assembled. The four geared DC motors are mounted at the corners of the four-wheel chassis, and the STM32 Blue Pill board, L298N motor driver, HC-05 Bluetooth module, and SSD1306 OLED display are secured on the upper section of the chassis. All wiring is properly connected and secured according to the circuit diagram. Fig. 6 shows the robot assembled on the chassis.

For testing, a 7.4V power supply is connected to the system. The L298N motor driver is tested by applying control signals from the STM32 Blue Pill to its IN1, IN2, IN3, and IN4 inputs. The motor responses are checked for forward, reverse, left-turn, right-turn, and 360-degree rotation.

Next, the HC-05 Bluetooth module is paired with a smartphone and different movement commands are sent through the mobile application. The robot's response to each command is verified. The SSD1306 OLED display is also tested by displaying startup messages, movement directions, operating status, and animations.

Finally, the fully assembled robot is tested with all modules operating together. Bluetooth communication, motor control, OLED display, and power supply are checked for reliable operation, and the robot is tested to ensure that it performs all required movements according to the design objectives. EFY

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